

Laboratory Practical Work 2

Quadrupole theory

Goals:

- Become familiar with resistive attenuator structures and constant impedance filters.
- Design adapted systems using concepts from classical theory.
- Use the concepts seen throughout the second semester of quadrupole theory, parameters Z, Y, T, Image and S, synthesis and analysis.

Approval condition: The practice must be completed in its entirety and submitted by email or via the campus within two weeks of the date of completion.

Responsible: Each practical assignment must have a person in charge.

Introduction

This practical laboratory exercise aims to design, simulate, implement and validate two circuits:

- Resistive attenuator.
- Constant impedance filter.

To complete this practical exercise, it is necessary to use several of the concepts seen throughout the second semester and to connect them by making two simple circuits like the ones presented.

Dissipative attenuator

A dissipative attenuator is to be designed. The design conditions for each group are:

Group	Attenuation [dB]	Z_{i1} [be]	Z_{i2} [be]
1	20	50	50
2	22	50	50
3	24	50	50
5	18	50	50

Synthesize the four-terminal network taking into account the error introduced when normalizing the components to commercial values so that the attenuation error can be kept within $\pm 1\text{dB}$. Consider using more than one stage.

Clarification: the attenuation is of voltage.

Instructions

- Calculate the components.
- Simulate and verify the design.
- Calculate the s parameters of the quadrupole.
- Implement the attenuator on a circuit board so that the laboratory day can be measured. Place terminals to connect the measuring instruments and the terminal load. **Do not weld the terminal load.**

Constant impedance filter

The filter to be implemented must comply with the following voltage transfer in the Laplace domain:

$$H_{(s)1} = \frac{V_2}{V} = \frac{s}{s^2 + s + 1}$$

This transfer is measured under load conditions for both port 1 and port 2 at 50Ω. For synthesis, a bridged T-type structure will be used. The center frequencies for each group are:

Group	f_0 [KHz]
1	900
2	1000
3	1100
5	1200

The circuit synthesis results in the construction of two coils of the same value. It is recommended to use the web application <http://www.pronine.ca/multind.htm> for the construction of the coils. The following rolls of wire are available in the department's storeroom for any group that needs them.



You can use a pen or a wick as the body of the coil and wind it to maintain the internal diameter.

It is advisable to choose an internal diameter close to 10mm and a length no greater than 30mm. Making double-layer wound inductors is not recommended to avoid errors in the winding direction.

To measure the components they manufacture, they can use the LCR Meter, which is shown below:



Clarification: The coils must be made prior to the laboratory day.

Instructions

- Calculate the components of the bridged T-type quadrupole.
- Simulate and verify the computed transfer.
- Implement the filter on a circuit board so that the laboratory day can be measured. Place terminals to connect the measuring instruments and the terminal load. **Do not weld the terminal load.**

Laboratory measurements

Part A

Ten data points should be taken across the frequency range of the function generator, and the magnitude response of the attenuator loaded to Z_o should be validated. Explain any differences compared to the simulation.

Part B

The function generator must be connected to the filter and its terminal load Z_o . Test at different frequencies and plot the frequency response. Measure between 20 and 30 points, appropriately spaced for visualization.

- Continuous behavior.
- Behavior close to the 3dB cutoff frequency.
- Center frequency.
- Behavior above the upper 3B cutoff frequency.

Graph and compare the obtained frequency response with the simulated response. Explain any differences.

Part C

Load the filter with an impedance approximately 10 times larger than Z_o and remeasure around the passband at approximately 10 points. Repeat the same procedure with a Z_o approximately 10 times smaller.

Graph and compare with the simulation. Explain what is happening.

Part D

Insert the designed attenuator between the generator and the filter, reload with Z_o , and validate the frequency characteristic. Explain the results obtained.

Remove the terminal load Z_o and place a load 10 times larger, then 10 times smaller. Analyze again what happens in the passband and compare with the results of the *Part C*.